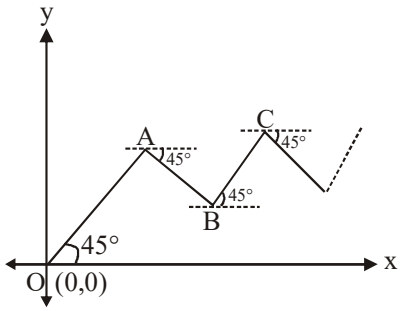
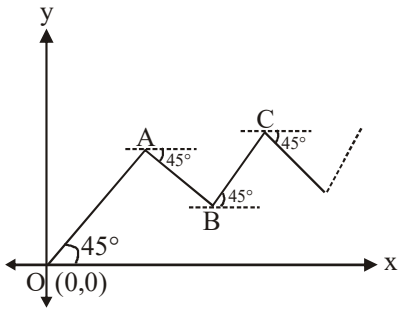
Exercise C - II

SUBJECTIVE / INTEGER TYPE QUESTIONS

1. Points O, A, B, C.....are shown in figure where OA = 2AB = 4BC = ... so on. Let A is the centroid of a triangle whose orthocentre and circumcenter are (2, 4) and $\left(\frac{7}{2}, \frac{5}{2}\right)$ respectively. If an insect starts moving from the point O(0, 0) along the straight line in zig–zag fashion and terminates ultimately at point P(α, β) then find the value of ($\alpha + \beta$)



आरेख में प्रदर्शित बिन्दुओं O, A, B, C ..... जहाँ OA = 2AB = 4BC = ... यदि बिन्दु A, किसी त्रिभुज का केन्द्रक है एवं जिसका लम्बकेन्द्र (2, 4) एवं परिकेन्द्र $\left(\frac{7}{2}, \frac{5}{2}\right)$ पर स्थित है, यदि कोई कीट मूल बिन्दु O(0, 0) से रेखा के साथ उतार–चढ़ाव (zig–zag) में बिन्दु P तक चलता है। यदि बिन्दु P के निर्देशांक (α, β) हो तो ($\alpha + \beta$) का मान ज्ञात करो।



Ans. 8

Sol. $H(2,4) \bullet \underset{G}{\overset{2 \quad 1}{\longrightarrow}} \bullet 0\left(\frac{7}{2}, \frac{5}{2}\right)$

$$\left(\frac{7+2}{+3}, \frac{5+4}{3}\right) \equiv G(3,3)$$

$$\alpha = \frac{3\sqrt{2}}{\sqrt{2}} + \frac{3\sqrt{2}}{2\sqrt{2}} + \frac{3\sqrt{2}}{4\sqrt{2}} + ...\infty = \frac{\frac{3\sqrt{2}}{\sqrt{2}}}{1-\frac{1}{2}} = 6$$

$$\beta = \frac{3\sqrt{2}}{\sqrt{2}} - \frac{3\sqrt{2}}{2\sqrt{2}} + \frac{3\sqrt{2}}{4\sqrt{2}} .....\infty = \frac{\frac{3\sqrt{2}}{\sqrt{2}}}{\frac{3}{2}} = 2$$

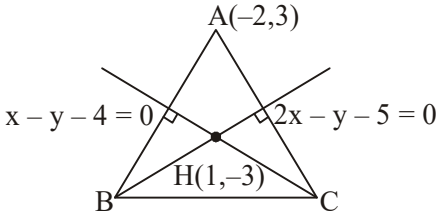
$\alpha + \beta = 6 + 2 = 8$

2. The equations of perpendiculars of the sides AB & AC of triangle ABC are $x - y - 4 = 0$ and $2x - y - 5 = 0$ respectively. If the vertex A is $(-2, 3)$ and point of intersection of perpendiculars bisectors is $\left(\frac{3}{2}, \frac{5}{2}\right)$, find the equation of medians to the sides AB & AC respectively.

त्रिभुज ABC समें भुजाओं AB एवं AC के लम्बो का समीकरण क्रमशः $x - y - 4 = 0$ एवं $2x - y - 5 = 0$ हो, यदि शीर्ष बिन्दु A के निर्देशांक $(-2, 3)$ हो, एवं $\left(\frac{3}{2}, \frac{5}{2}\right)$ भुजाओं के लम्ब अर्द्धको का प्रतिच्छेद बिन्दु हो, तो क्रमशः AB एवं AC की मध्यिकाओं के समीकरण ज्ञात करो।

Ans. $x + 4y = 4$; $5x + 2y = 8$

Sol.



$$\begin{array}{rrrrrr} x & - & y & - & 4 & = & 0 \\ 2x & - & y & - & 5 & = & 0 \\ - & & + & & + & & \\ \hline & - & x & + & 1 & = & 0 \\ \hline \end{array}$$

$x = 1 \qquad y = -3$

Eqn of $\perp$ bisector of AB and AC $\rightarrow$ AB $\perp^r$ Bisector $\rightarrow x - y + 1 = 0$

$x - y + 1 = 0$

AC $\perp^r$ Bisector

$$2x - y - \frac{1}{2} = 0$$

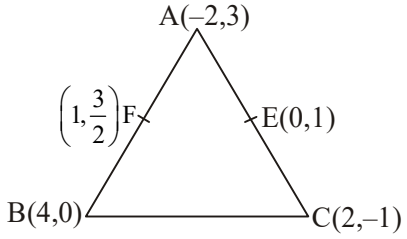
$$4x - 2y - 1 = 0$$

$B \rightarrow$ Image of $A(-2, 3)$ in $4x - 2y - 1 = 0$

$$\frac{x+2}{4} = \frac{y-3}{-2} = -2\frac{(-8-6-1)}{20} = \frac{3}{2} \Rightarrow B(4,0)$$

$C \rightarrow$ Image of $A(-2, 3)$ in $x - y + 1 = 0$

$$\frac{x+2}{1} = \frac{y-3}{(-1)} = -2\frac{(-2-3+1)}{2} = 4 \Rightarrow C(2,-1)$$



CF : $\rightarrow$ $\quad (y+1) = \left(\frac{\frac{3}{2}+1}{1-2}\right)(x-2)$

$$2y + 2 = -5x + 10$$

$$5x + 2y = 8 \text{ Ans.}$$

BE : $\rightarrow$ $\quad (y-0) = \frac{1}{-4}(x-4)$

$$-4y = x - 4$$

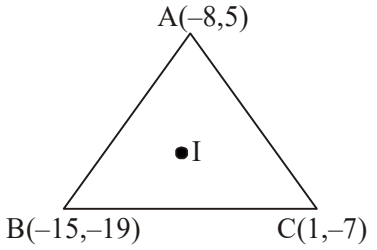
$$x + 4y = 4 \text{ Ans.}$$

3. The interior angle bisector of angle A for the triangle ABC whose coordinates of the vertices are A(–8, 5); B(–15, –19) and C(1, – 7) has the equation $ax + 2y + c = 0$. Find 'a' and 'c'.

यदि त्रिभुज ABC के शीर्ष बिन्दु के निर्देशांक A(–8, 5); B(–15, –19) एवं C(1, – 7) है, एवं कोण के आंतरिक कोण अर्द्धक का समीकरण $ax + 2y + c = 0$ है तो 'a' एवं 'c' का मान ज्ञात करो।

Ans. $a = 11$, $c = 78$

Sol.



$I \equiv$ Incentre

$a = BC = \sqrt{16^2 + 12^2}$

$a = 20$

$b = AC = \sqrt{9^2 + 12^2}$

$b = 15$

$C = AB = \sqrt{7^2 + 24^2}$

$C = 25$

$$I = \left(\frac{-160 - 225 + 25}{60}, \frac{100 - 285 - 175}{60} \right) \qquad \left(\because I \equiv \frac{ax_1 + by_1 + cz_1}{a+b+c}, \frac{ax_2 + by_2 + cz_2}{a+b+c} \right)$$

$I \equiv (-6, -6)$

AI eqn. : $\rightarrow \quad (y+6) = \frac{11}{-2}(x+6)$

$-2y - 12 = 11x + 66$

$11x + 2y + 78 = 0$

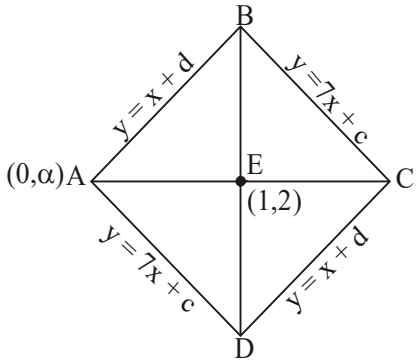
$a = 11,\ c = 78$

4. Two sides of a rhombous ABCD are parallel to the lines $y = x + 2$ & $y = 7x + 3$. If the diagonals of the rhombous intersect at the point (1, 2) & the vertex A is on the y-axis, find the possible coordinates of A.

किसी समचतुर्भुज ABCD की दो भुजाऐं, रेखाओं $y = x + 2$ एवं $y = 7x + 3$ के समान्तर हो, एवं समचतुर्भुज के विकर्णो का प्रतिच्छेद बिन्दु (1, 2) हो तथा बिन्दु A, y-अक्ष पर स्थित हो, तो बिन्दु A के सम्भावित निर्देशांक ज्ञात करो।

Ans. $(0, 0)$ or $\left(0, \frac{5}{2}\right)$

Sol.



Diagonals will be parallel to angle bisectors of line

$y = x + 2$

$y = 7x + 3$

$$\frac{x - y + 2}{\sqrt{2}} = \pm\frac{(7x - y + 3)}{5\sqrt{2}}$$

(+)	(–)
$5x - 5y + 10 = 7x - y + 3$	$5x - 5y + 10 = -7x + y - 3$
$B_1 : 2x + 4y - 7 = 0$	$B_2 : 12x - 6y + 13 = 0$
$x + 2y - \frac{7}{2} = 0$	$2x - y + \frac{13}{6} = 0$

$$m_{AE} = \frac{-2}{4} \text{ or } \frac{12}{6}$$

$$\Rightarrow \frac{2-\alpha}{1-0} = -\frac{1}{2} \text{ or } \frac{2-\alpha}{1-0} = 2$$

$$\alpha = \frac{5}{2} \text{ or } \alpha = 0$$

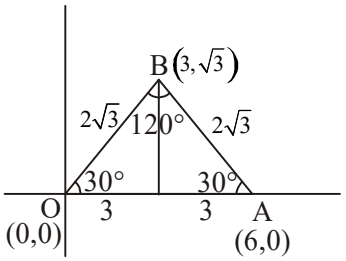
Coordinates $\left(0, \frac{5}{2}\right)$ or $(0,0)$

5. Let O(0, 0), A(6, 0) and B(3, $\sqrt{3}$) be the vertices of ΔOAB. Let R be the region consisting of all those points P inside ΔOAB which satisfy d(P, OA) ≤ minimum $\left(d(P, OB), d(P, AB)\right)$ where d(P, OA), d(P, OB) and d(P, AB) represent the distance of P from the sides OA, OB and AB respectively. If the area of region R is 9 ($a - \sqrt{b}$) where a and b are coprime, then find the value of (a + b).

यदि O(0, 0), A (6, 0) एवं B(3, $\sqrt{3}$) त्रिभुज OAB के शीर्ष बिन्दु है तथा R त्रिभुज OAB के अंदर स्थित सभी बिन्दुओं के क्षेत्र को प्रदर्शित करता है जो कि संबंध d(P, OA) ≤ न्यूनतम मान $\left(d(P,OB), d(P,AB)\right)$ को संतुष्ट करता है जहाँ d(P, OA), d(P, OB) तथा d(P, AB) बिन्दु क्रमशः भुजाओं OA, OB एवं AB दूरी को दर्शाता है। यदि क्षेत्र R का क्षेत्रफल 9 (a – $\sqrt{b}$) है जहाँ a एवं b परस्पर अभाज्य (coprime) है, तो (a + b) का मान ज्ञात कीजिये।

Ans. 5

Sol.



Let P(h, k)

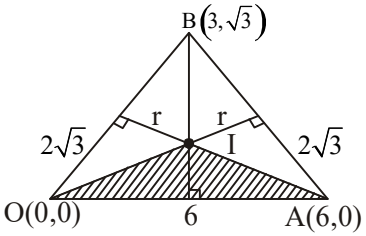
$OB \equiv \sqrt{3}y = x$

$AB = \sqrt{3}y = -x + 6$

$d(P,OA) \le \min\left\{d(P,OB), d(P,AB)\right\}$

$d(P,OA) \le d(P,OB)$

$d(P,OA) \le d(P,AB)$



$$I\left(\frac{18+0+12\sqrt{3}}{6+4\sqrt{3}}, \frac{6\sqrt{3}+0+0}{6+4\sqrt{3}}\right)$$

$$\text{Area} = R = \frac{1}{2}\times(6)\times\left(\frac{6\sqrt{3}}{6+4\sqrt{3}}\right)$$

$$= 9\left(\frac{\sqrt{3}}{3+2\sqrt{3}}\right) = 9\left(\frac{\sqrt{3}\left(2\sqrt{3}-3\right)}{3}\right)$$

$$= 9\left(2-\sqrt{3}\right)$$

a = 2, b = 3, a + b = 5

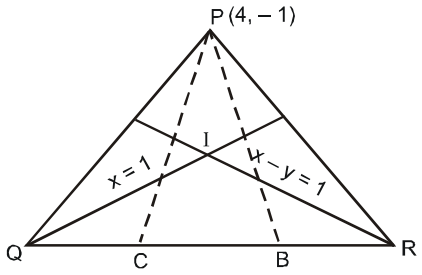
6. Find the equations of the sides of a triangle having (4, –1) as a vertex, if the lines x – 1 = 0 and x – y –1 = 0 are the equations of two internal bisectors of its angles.

त्रिभुज की भुजाओं के समीकरण ज्ञात करो यदि एक शीर्ष बिन्दु के निर्देशांक (4, –1) एवं दो आन्तरिक कोर्णाद्धको के समीकरण x – 1 = 0 एवं x – y –1 = 0 है।

Ans. 2x – y + 3 = 0, 2x + y – 7 = 0, x – 2y – 6 = 0

Sol. **Method I** :

Alter :



Mirror image of a vertex of ΔABC w.r.t. it's angle bisector always lie on the side opposite to the vertex.

Reflection of (4, – 1) about x = 1 is B and about x – y = 1 is C

$B \equiv (x_1, y_1)$

$C \equiv (x_2, y_2)$

$\left(\frac{x_1+4}{2}, \frac{y_1-1}{2}\right)$ will lie on x = 1

$\Rightarrow \quad \frac{x_1+4}{2} = 1 \quad \Rightarrow \quad x_1 = -2 \qquad$ and $\qquad \frac{y_1+1}{x_1-4} = 0 \quad \Rightarrow \quad y_1 = -1$

$\therefore \quad B \equiv (-2, -1) \Rightarrow \left(\frac{x_2+4}{2}, \frac{y_2-1}{2}\right)$ will lie on x – y – 1 = 0

$x_2 + 4 - y_2 + 1 - 2 = 0 \quad \Rightarrow \quad x_2 - y_2 + 3 = 0 \quad$ _____(i)

and $\left(\frac{y_2+1}{x_2-4}\right).1 = -1 \quad \Rightarrow \quad x_2 + y_2 - 3 = 0 \quad$ _____(ii)

from (i) & (ii) $x_2 = 0, y_2 = 3$

$\therefore$ C(0, 3) $\therefore$ Equation of BC

$y + 1 = \frac{4}{2}(x + 2) \quad \Rightarrow \quad 2x - y + 3 = 0$

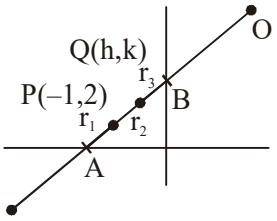
To find PQ & PR we have to find Q and R ;
solving QI & BC, we get Q (1, 5)
equation of PQ is $2x + y - 7 = 0$
For R, we solve RI & BC
$\therefore$ R (–4, – 5)
equation of PR is $x - 2y - 6 = 0$

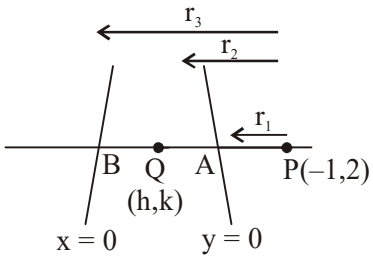
7. P is the point (–1, 2), a variable line through P cuts the x & y axes at A & B respectively Q is the point on AB such that PA, PQ, PB are H.P. Find the locus of Q.

यदि P के निर्देशांक (–1, 2) हो, एवं एक चर रेखा जो बिन्दु P से गुजरती है, तथा x एवं y अक्षों को क्रमशः A एवं B पर काटती है, एवं Q, AB पर स्थित कोई बिन्दु है, तथा PA, PQ, PB हरात्मक श्रेणी में हो तो, बिन्दु Q का बिन्दुपथ ज्ञात करो।

Ans. $y = 2x$

Sol.





$$Q(h,k) = Q(-1 + r_2\cos\theta, 2 + r_2\sin\theta)$$

$$A(-1 + r_1\cos\theta, 2 + r_1\sin\theta) \Rightarrow x - \text{axis}$$

$$r_1\sin\theta = -2$$

$$r_1 = \frac{-2}{\sin\theta}$$

$$B(-1 + r_3\cos\theta, 2 + r_3\sin\theta) \Rightarrow y - \text{axis}$$

$$-1 + r_3\cos\theta = 0$$

$$r_3 = \frac{1}{\cos\theta}$$

$$\frac{2}{r_2} = \frac{1}{r_1} + \frac{1}{r_3} = \frac{-\sin\theta}{2} + \frac{\cos\theta}{1}$$

$$2 = \frac{-r_2 \sin\theta}{2} - r_2 \cos\theta$$

$$4 = -(k-2) + 2(h+1)$$

$2h - k = 0$

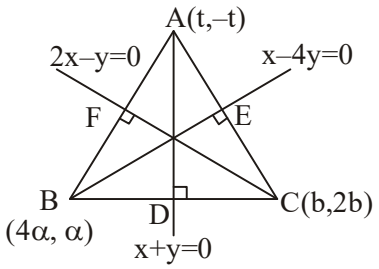
$y = 2x$

8. The equations of the altitudes AD, BE, CF of a triangle ABC are $x+y=0$, $x-4y=0$ and $2x-y=0$ respectively. The coordinates of A are $(t, -t)$. Find coordinates of B & C. Prove that if t varies the locus of the centroid of the triangle ABC is $x+5y=0$.

त्रिभुज ABC के लम्बो AD, BE, CF के समीकरण के क्रमशः $x+y=0$, $x-4y=0$ एवं $2x-y=0$ है एवं बिन्दु A के निर्देशांक $(t, -t)$ है। तो बिन्दु B एवं C के निर्देशांक ज्ञात कीजिये। दर्शाओं यदि t प्राचल हो तो त्रिभुज ABC के केन्द्रक का बिन्दुपथ $x+5y=0$ होगा।

Ans. $B\left(-\frac{2t}{3}, -\frac{t}{6}\right)$, $C\left(\frac{t}{2}, t\right)$

Sol.



$$\frac{-t-\alpha}{t-4\alpha} = \frac{-1}{2}$$

$2t + 2\alpha = t - 4\alpha$

$t = -6\alpha$

$$\alpha = \frac{t}{-6}$$

$$\frac{2b+t}{b-t} = -4$$

$2b + t = 4t - 4b$

$6b = 3t$

$t = 2b$

$$b = \frac{t}{2}$$

$$B\left(\frac{-2t}{3}, \frac{-t}{6}\right) \qquad C\left(\frac{t}{2}, t\right)$$

$$G\left(\frac{t - \frac{2t}{3} + \frac{t}{2}}{3}, \frac{-t - \frac{t}{6} + t}{3}\right) \equiv (h, k)$$

$$\left.\begin{array}{l} \frac{5t}{18} = h \Rightarrow t = \frac{18h}{5} \\ t = -18k \end{array}\right\rangle \quad h + 5k = 0$$

$$\Rightarrow x + 5y = 0$$

hence, locus of G is st. line $x + 5y = 0$

9. The distance of a point (x_1, y_1) from each of two straight lines which passes through the origin of co-ordinates is δ; find the combined equation of these straight lines.

बिन्दु P (x_1, y_1) की मूल बिन्दु से गुजरने वाली दो रेखाओं में प्रत्येक दूरी δ है। तो दोनो रेखाओं का सम्मिलित समीकरण ज्ञात करो।

Ans. $(y_1^2 - \delta^2)\, x^2 - 2\, x_1 y_1\, xy + (x_1^2 - \delta^2)\, y^2 = 0$

Sol. $$P_1\, P_2 = \frac{\left|ax_1^2 + 2hx_1y_1 + by_1^2\right|}{\sqrt{(a-b)^2 + 4h^2}}$$

$$\delta^2 = \frac{\left|ax_1^2 + 2hx_1y_1 + by_1^2\right|}{\sqrt{(a-b)^2 + 4h^2}}$$

$y = m_1x, \qquad y = m_2x$

$$\delta = \left|\frac{m_1x_1 - y_1}{\sqrt{1 + m_1^2}}\right| \qquad \delta = \left|\frac{m_2x_2 - y_2}{\sqrt{1 + m_2^2}}\right|$$

$$\delta^2 = \frac{(mx_1 - y_1)^2}{1 + m_1^2} \Rightarrow \delta^2 = \frac{m^2x_1^2 - 2x_1y_1m + y_1^2}{1 + m_1^2}$$

$$\left(x_1^2 - \delta^2\right)m^2 - 2x_1y_1m + \left(y_1^2 - \delta^2\right) = 0$$

Put $m = \frac{y}{x}$

$\left(y_1^2 - \delta^2\right)x^2 - 2x_1y_1xy + \left(x_1^2 - \delta^2\right)xy^2 = 0$

10. The sides of a triangle have the combined equation $x^2 - 3y^2 - 2xy + 8y - 4 = 0$. The third side, which is variable always passes through the point (–5, –1). If the range of values of the slope of the third line so that the origin is an interior point of the triangle, lies in the interval (a, b) then find $\left(a + \frac{1}{b^2}\right)$.

त्रिभुज की दो भुजाओं का संयुक्त रूप का समीकरण $x^2 - 3y^2 - 2xy + 8y - 4 = 0$ तथा तीसरी चर भुजा जो कि हमेशा बिन्दु (–5, –1) से गुजरती है। तीसरी भुजा की प्रवणता का अन्तराल (a, b) इस तरह है, कि मूल बिन्दु त्रिभुज का आंतरिक बिन्दु है तो $\left(a + \frac{1}{b^2}\right)$ का मान ज्ञात करो।

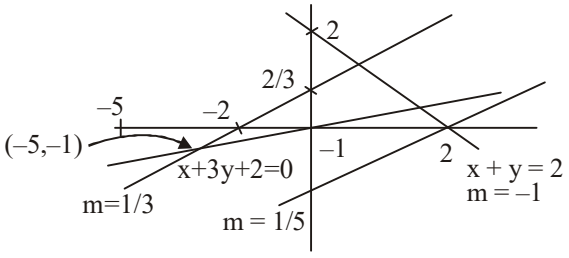
Ans. 24

Sol. $x^2 - 2xy - 3y^2 + 8y - 4 = 0$

$x^2 - 2xy - 3y^2 = x^2 - 3xy + xy - 3y^2 = (x + y)(x - 3y) = 0$

$(x + y + c_1)(x - 3y + c_2) = 0$

$$\begin{array}{rrrrr} & C_1 & + & C_2 & = 0 \\ - & 3C_1 & + & C_2 & = 8 \\ + & & - & & - \\ \hline & & 4C_1 & = & -8 \\ \hline \end{array}$$



$C_1\, C_2 = -4$

$C_1 = -2 \qquad C_2 = 2$

$L_1 : x + y - 2 = 0 \qquad L_2 : x - 3y + 2 = 0$

Let equation of variable line be

$y + 1 = m(x + 5)$

For origin to lie inside the required Δ, y intercept for the variable line should be –ve.

$\therefore\ 5m - 1 = 0 \Rightarrow m < \frac{1}{5}$

$(-5, -1)$ also lie on $x - 3y + 2 = 0$

Now, Δ can form when $m > -1$.

So, $m \in \left(-1, \frac{1}{5}\right)$, $a + b = 4$

11. Consider a line pair $2x^2 + 3xy - 2y^2 - 10x + 15y - 28 = 0$ and another line L passing through origin with gradient 3. The line pair and line L form a triangle whose vertices are A, B and C.

(a) Find the sum of the contangents of the interior angles of the triangle ABC.

(b) Find the area of triangle ABC

(c) Find the radius of the circle touching all the 3 sides of the triangle.

एक रेखा युग्म $2x^2 + 3xy - 2y^2 - 10x + 15y - 28 = 0$ तथा रेखा L जिसकी प्रवणता 3 एवं मूलबिन्दु से गुजरती है। तो रेखा युग्म एवं रेखा L से बनने वाले त्रिभुज के शीर्ष बिन्दु A, B एवं C है।

(a) CotA + CotB + CotC का मान ज्ञात कीजिये। (जहाँ A,B,C त्रिभुज ABC के आन्तरिक कोण हैं)

(b) त्रिभुज ABC का क्षेत्रफल ज्ञात करो

(c) त्रिभुज की तीनो भुजाओं को स्पर्श करने वाले वृत की त्रिज्या ज्ञात करे।

Ans. (a) $\frac{50}{7}$; (b) $\frac{63}{10}$; (c) $\frac{3}{10}\left(8\sqrt{5} - 5\sqrt{10}\right)$

Sol. $2x^2 + 3xy - 2y^2 - 2y^2 - 10x + 15y - 28 = 0$

$y = 3x$

$2x^2 + 3xy - 2y^2 = 0$

$2 + 3m - 2m^2 = 0 \qquad \left(\because \frac{y}{x} = m\right)$

$2m^2 - 3m - 2 = 0$

$2m^2 - 4m + m - 2 = 0$

$(2m + 1)(m - 2) = 0$

$m = -\frac{1}{2}, \qquad m = 2$

$m_1 = 3$

$m_2 = 2$

$$m_3 = -\frac{1}{2}$$

$$\tan I = \frac{m_1 - m_2}{1 + m_1 m_2} = \frac{3-2}{1+6} = \frac{1}{7}$$

$$\tan II = \frac{m_2 - m_3}{1 + m_2 m_3} = \frac{2 + \frac{1}{2}}{1-1} = \infty$$

$$\tan III = \frac{-\frac{1}{2} - 3}{1 - \frac{3}{2}} = \frac{-17}{1+1}$$

(a) $\sum \cot I = 7 + \frac{1}{7} + 0 = \frac{50}{7}$

(d) $2x^2 + 3xy - 2y^2 = 0$

$2x^2 + 4xy - xy - 2y^2 = 0$

$2x(2x + 2y) - y(x + 2y) = 0$

$(x + 2y)(2x - y) = 0$

$2C_1 + C_2 = -10$ $\qquad C_1 C_2 = -28$

$-C_1 + 2C_2 = 15$

$C_1 = -7 \qquad C_2 = 4$

$x + 2y + C_1 = 0 \qquad \Rightarrow x + 2y - 7 = 0$ and $y = 3x$

$2x - y + C_2 = 0 \qquad \Rightarrow 2x - y + 4 = 0$

P.O.I $\quad A\left(-\frac{1}{5}, \frac{18}{5}\right)$

B (4, 12)

C (1, 3)

$$\Delta ABC = \frac{1}{2}\begin{vmatrix} -\frac{1}{5} & \frac{18}{5} & 1 \\ 4 & 12 & 1 \\ 1 & 3 & 1 \end{vmatrix} = \frac{1}{10}\begin{vmatrix} -1 & 18 & 5 \\ 4 & 12 & 1 \\ 1 & 3 & 1 \end{vmatrix}$$

$$= \frac{1}{10}\begin{vmatrix} -6 & 3 & 5 \\ 3 & 9 & 1 \\ 0 & 0 & 1 \end{vmatrix} \Rightarrow \frac{1}{10}[-54-9] = \frac{63}{10}$$

12. The two line pairs $y^2 - 4y + 3 = 0$ and $x^2 + 4xy + 4y^2 - 5x - 10y + 4 = 0$ enclose a 4 sided convex polygon find
(i) area of the polygon (ii) length of its diagonals.

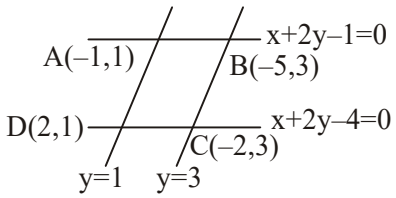
दो रेखा युग्मों $y^2 - 4y + 3 = 0$ एवं $x^2 + 4xy + 4y^2 - 5x - 10y + 4 = 0$ से चार भुजाओं वाला उत्तल बहुभुज निर्मित होता है, तो
(i) बहुभुज का क्षेत्रफल (ii) विकर्णो की लम्बाई ज्ञात करो

Ans. (i) area = 6 sq. units, (ii) diagonals are $\sqrt{5}$ & $\sqrt{53}$

Sol. $y^2 - 4y + 3 = 0$ $\qquad x^2 + 4xy + 4y^2 - 5x - 10y + 4 = 0$

$y = 3, 1$ $\qquad (x + 2y)^2 - 5(x + 2y) + 4 = 0$

$(x + 2y - 4)(x + 2y - 1) = 0$



(i) Area of ploygon

$$\text{Area} = \frac{1}{2}\begin{vmatrix} -1 & 2 & -2 & -5 & -1 \\ 1 & 1 & 3 & 3 & 1 \end{vmatrix}$$

$$= \frac{1}{2}[-1 - 2 + 6 + 2 - 6 + 15 - 5 + 3]$$

$= 6$

(ii) Length of diagonals

$AC = \sqrt{1+4} = \sqrt{5}$

$BD = \sqrt{49+4} = \sqrt{53}$

13. Find the equation of the two straight lines which together with those given by the equation $6x^2 - xy - y^2 + x + 12y - 35 = 0$ will make a parallelogram whose diagonals intersect in the origin.

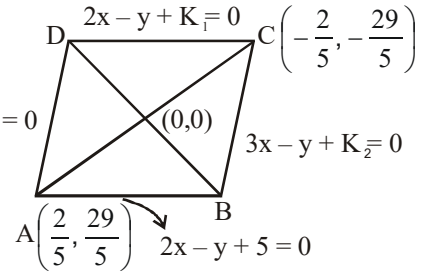
उन दोनों रेखाओं का समीकरण ज्ञात करो जो कि समीकरण $6x^2 - xy - y^2 + x + 12y - 35 = 0$ से प्रदर्शित होने वाली अन्य रेखाओं के साथ मिलकर एक सामान्तर चतुर्भुज इस प्रकार निर्मित करती है कि समान्तर चतुर्भुज के विकर्ण मूलबिन्दु पर प्रतिच्छेद करते है।

Ans. $6x^2 - xy - y^2 - x - 12y - 35 = 0$

Sol. $6x^2 - xy - y^2 + x + 12y - 35 = 0 \quad \Rightarrow \quad (2x - y + 5)(3x + y - 7) = 0$

Now point A is intersection point of $2x - y + 5 = 0$ and $3x + y - 7$

$\therefore \quad A\left(\frac{2}{5}, \frac{29}{5}\right)$



Now C $\left(-\frac{2}{5}, -\frac{29}{5}\right)$ as midpoint of diagonals is (0, 0)

$2x - y + K_1 = 0$ and $3x + y + K_2 = 0$ are passing through C

$\therefore \quad K_1 = -5$, and $K_2 = 7$

Now combined equation $(2x - y - 5)(3x + y + 7) = 0$

$6x^2 - xy - y^2 - x - 12y - 35 = 0$

14. A straight line passing through O(0, 0) cuts the lines $x = a$, $y = b$ and $x + y = 8$ at A, B and C respectively such that $OA \cdot OB \cdot OC = 48\sqrt{2}$ and $f(a, b) \le 0$

where $f(x, y) = \left|\frac{y}{x} - \frac{3}{2}\right| + (3x - 2y)^6 + \sqrt{ex + 2y - 2e - 6}$.

(i) Find the point of intersection of lines $x = a$ and $y = b$.

(ii) Find the value of $(OA + OB + OC)$.

(iii) Find the equation of line OA.

एक सरल रेखा जो कि मूल बिन्दु O(0, 0) से गुजरती है, तथा रेखाओं $x = a$, $y = b$ एवं $x + y = 8$ को क्रमशः A, B एवं C पर इस प्रकार प्रतिच्छेद करती है, कि $OA \cdot OB \cdot OC = 48\sqrt{2}$ एवं $f(a, b) \le 0$

जहाँ $f(x, y) = \left|\frac{y}{x} - \frac{3}{2}\right| + (3x - 2y)^6 + \sqrt{ex + 2y - 2e - 6}$

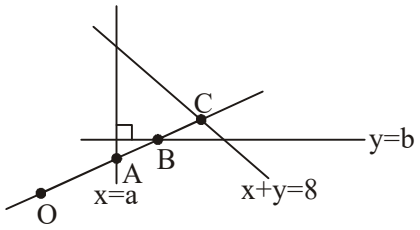
(i) रेखाओं $x = a$ एवं $y = b$ का प्रतिच्छेद बिन्दु ज्ञात करो

(ii) $(OA + OB + OC)$ का मान ज्ञात करो।

(iii) OA का समीकरण ज्ञात करो।

Ans. (i) a = 2 and b = 3; (ii) $9\sqrt{2}$; (iii) x – y = 0

Sol.



(i) $f(a, b) \le 0$

$\Rightarrow f(a, b) = 0$ (only possible)

$\frac{b}{a} = \frac{3}{2}$

$2b = 3a$

$ea + 2b - ea - 6 = 0$

$(a, b) = (2, 3)$ $\quad b = 3$

$\quad a = 2$

(ii) $A\left(r_1 \cos\theta, r_1 \sin\theta\right)$ $\quad \Rightarrow r_1 \cos\theta = 2$

$B\left(r_2 \cos\theta, r_2 \sin\theta\right)$ $\quad \Rightarrow r_2 \sin\theta = 3$

$C\left(r_3 \cos\theta, r_3 \sin\theta\right)$ $\quad \Rightarrow r_3 = \frac{8}{\cos\theta + \sin\theta}$

$r_1 r_2 r_3 = 48\sqrt{2}$

$\frac{2 \times 3 \times 8}{\cos\theta.\sin\theta.\left(\cos\theta + \sin\theta\right)} = 48\sqrt{2}$ $\quad \Rightarrow \sin\theta\cos\theta\left(\sin\theta + \cos\theta\right) = \frac{1}{\sqrt{2}}$

$\Rightarrow \sin 2\theta.\left(\sin\theta + \cos\theta\right) = \sqrt{2}$

$\sin 2\theta = 1,$ $\quad \sin\theta + \cos\theta = \sqrt{2}$

$\theta = 45°$

$r_1 = 2\sqrt{2}$, $r_2 = 3\sqrt{2}$, $r_3 = \frac{8}{2}\sqrt{2} = 4\sqrt{2}$

$r_1 + r_2 + r_3 = 9\sqrt{2}$

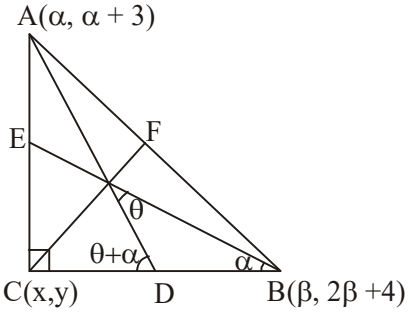
(iii) $y = x$ $\quad m = \tan 45° = 1,$ $\quad O(0,0)$

15. The triangle ABC, right angled at C, has median AD, BE and CF. AD lies along the line $y = x + 3$, BE lies along the line $y = 2x + 4$. If the length of the hypotenuse is 60, find the area of the triangle ABC.

त्रिभुज ABC, जिसक कोण C समकोण है, तथा माध्यिकाये AD, BE एवं CF है जहाँ AD एवं BE क्रमशः रेखाओं $y = x + 3$ एवं $y = 2x + 4$ के अनुसरित है यदि कर्ण की लम्बाई 60 है तो त्रिभुज ABC का क्षेत्रफल ज्ञात करो।

Ans. 400 sq. units

Sol.



$AC^2 + BC^2 = 60^2$

$b^2 + a^2 = 60^2$

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right| = \left|\frac{1-2}{1+2}\right| = \frac{1}{3}$$

In ΔCBE

$$\tan\theta = \frac{CE}{BC} = \frac{\frac{b}{2}}{a} = \frac{b}{2a}$$

In ΔADC

$$\tan(\theta + \alpha) = \frac{AC}{CD} = \frac{b}{\frac{a}{2}} = \frac{2b}{a}$$

$$\frac{\tan\theta + \tan\alpha}{1 - \tan\theta\tan\alpha} = \frac{2b}{a} \Rightarrow \frac{\frac{1}{3} + \frac{b}{2a}}{1 - \frac{1}{3}\cdot\frac{b}{2a}} = \frac{2b}{a}$$

$$\Rightarrow \frac{2a + 3b}{6a - b} = \frac{2b}{a}$$

$\Rightarrow ab = 800$

$ar.(\Delta ABC) = 400$

16. A straight line L through the origin meets the line $x+y=1$ and $x+y=3$ at P and Q respectively. Through P and Q two straight lines L_1 and L_2 are drawn, parallel to $2x-y=5$ and $3x+y=5$ respectively. Lines L_1 and L_2 intersect at R. Show that the locus of R, as L varies, is a straight line.

मूलबिन्दु से होकर जाने वाली एक सरल रेखा L रेखाओं $x+y=1$ और $x+y=3$ से क्रमशः P और Q पर मिलती है। P और Q से दो सरल रेखाऐं L_1 एवं L_2 क्रमशः $2x-y=5$ और $3x+y=5$ के समान्तर खींची जाती है। रेखाऐं L_1 व L_2 बिन्दु R पर प्रतिच्छेद करती है। दिखाइए कि बिन्दु R का बिन्दुपथ एक सरलरेखा है।

Ans. $x-3y+5=0$;

Sol. The line $y=mx$ meets the given lines in

$$P\left(\frac{1}{m+1},\frac{m}{m+1}\right) \text{ and } Q\left(\frac{3}{m+1},\frac{3m}{m+1}\right)$$

Hence equation of L_1 is

$$y-\frac{m}{m+1}=2\left(x-\frac{1}{m+1}\right)$$

$$\Rightarrow y-2x-1=-\frac{3}{m+1} \qquad \text{_____(1)}$$

and that of L_2 is

$$y-\frac{3m}{m+1}=-3\left(x-\frac{3}{m+1}\right)$$

$$\Rightarrow y+3x-3=\frac{6}{m+1} \qquad \text{_____(2)}$$

From (1) and (2) $\frac{y-2x-1}{y+3x-3}=-\frac{1}{2}$

$\Rightarrow x-3y+5=0$ which is a straight line

17. A straight line L with negative slope passes through the point (8,2) and cuts the positive coordinates axes at points P and Q. Find the absolute minimum value of OP + OQ, as L varies, where O is the origin.

ऋणात्मक प्रवणता वाली सरलरेखा L बिन्दु (8,2) से गुजरती है तथा निर्देश अक्षों का धनात्मक दिशा में P तथा Q बिन्दुओं पर काटती है। L के परिवर्तित होने के अनुसार OP + OQ का न्यूनतम निरपेक्ष मान ज्ञात कीजिये, जहाँ O मूल बिन्दु है।

Ans. 18

Sol. Equation of line

(A) $y-2=m(x-8)$ and $m<0 \quad \Rightarrow \quad P\left(8-\frac{2}{m},\ 0\right)$ and $Q(0,\ 2-8m)$

$$OP+OQ=\left|8-\frac{2}{m}\right|+|2-8m|=10+\frac{2}{(-m)}+8(-m) \geq 10+2\sqrt{\frac{2}{(-m)}\times 8(-m)} \geq 18.$$

Area of ΔOPQ is minimum when (8,2) is midpoint of line. So, P(16, 0), Q(0,4) $\Delta OPQ = \frac{1}{2}(16)(4) = 32$.

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 18 to 20

Let ABCD is a square with sides of unit length. Points E and F are taken on sides AB and AD respectively so that AE = AF. Let P be a point inside the square ABCD.

माना ABCD एक वर्ग है जिसकी भुजाओं की लम्बाई एक है। बिन्दु E तथा F क्रमशः भुजाओं AB तथा AD पर इस प्रकार लिये गये है कि AE = AF है। माना बिन्दु P वर्ग ABCD के अन्दर है।

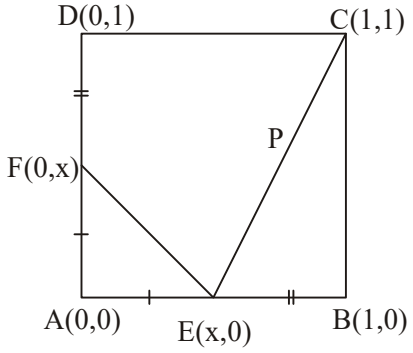
18. The maximum possible area of quadrilateral CDFE is :

चतुर्भुज CDFE का अधिकतम क्षेत्रफल है :

(A) $\frac{1}{8}$ (B) $\frac{1}{4}$ (C) $\frac{5}{8}$ (D) $\frac{3}{8}$

Ans. C

Sol.



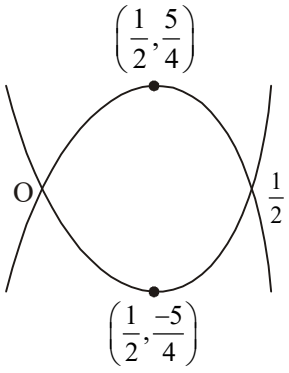
$$\text{Area} = \frac{1}{2}\begin{vmatrix} x & 0 & 0 & 1 & x \\ 0 & x & 1 & 1 & 0 \end{vmatrix}$$

$$A = \frac{1}{2}\left|x^2 - 1 - x\right|$$

$$A = \frac{1}{2}\left|x^2 - x - 1\right|$$

$$x \in (0,1)$$

$$\frac{-b}{2a} = \frac{1}{2}$$



$$\frac{-D}{4a}=\frac{-(1+4)}{4}=\frac{-5}{4}$$

$$A_{max}=\frac{5}{8}$$

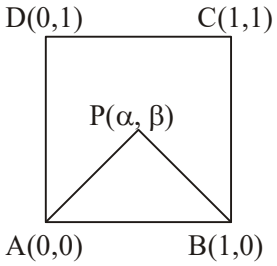
19. The value of $(PA)^2-(PB)^2+(PC)^2-(PD)^2$ is equal to :

$(PA)^2-(PB)^2+(PC)^2-(PD)^2$ का मान है :

(A) 3 (B) 2 (C) 1 (D) 0

Ans. D

Sol.



$$Z=\alpha^2+\beta^2-(\alpha-1)^2-\beta^2+(\alpha-1)^2+(\beta+2)^2-\alpha^2-(\beta\alpha)^2$$

$Z=0$

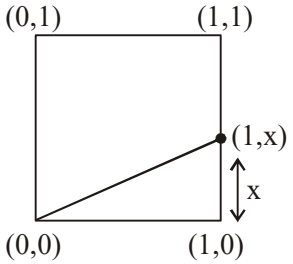
20. Let a line passing through point A divides the square ABCD into two parts so that area of one portion is double the other, then the length of portion of line inside the square is :

माना एक रेखा बिन्दु A से गुजरती है जो वर्ग ABCD को दो भागों में विभाजित करती है जिनमें एक का क्षेत्रफल अन्य से दोगुना है तो रेखा के वर्ग के भीतर हिस्से की लम्बाई है :

(A) $\frac{\sqrt{10}}{3}$ (B) $\frac{\sqrt{13}}{3}$ (C) $\frac{\sqrt{11}}{3}$ (D) $\frac{2}{\sqrt{3}}$

Ans. B

Sol.



$3A=1$

$$A = \frac{1}{3}$$

$$\frac{1}{2} \times 1 \times x = \frac{1}{3}$$

$$x = \frac{2}{3}$$

$$\text{Length} = \sqrt{1 + x^2}$$

$$= \sqrt{1 + \frac{4}{9}} = \frac{\sqrt{13}}{3}$$

Paragraph for Question Nos. 21 to 23

The base of an isosceles triangle is equal to 4, the base angle is equal to 45°. A straight line cuts the extension of the base at a point M at the angle θ and bisects the lateral side of the triangle which is nearest to M.

एक समद्विबाहु त्रिभुज का आधार 4 है तथा आधार कोण 45° है एक रेखा इस बढ़े हुये आधार को बिन्दु M पर θ कोण पर काटती है तथा M से करीब भुजा को समद्विभाजित करती है।

21. The area 'A' of the quadrilateral which the straight line cuts off from given triangle is :

यह रेखा त्रिभुज के साथ जो चतुर्भुज बनाती है। उसका क्षेत्रफल 'A' होगा :

(A) $\frac{3 + \tan\theta}{1 + \tan\theta}$ (B) $\frac{3 + 2\tan\theta}{1 + \tan\theta}$ (C) $\frac{3 + \tan\theta}{1 - \tan\theta}$ (D) $\frac{3 + 5\tan\theta}{1 + \tan\theta}$

Ans. D

22. The range of values of 'A' for different values of θ, lie in the interval :

θ के भिन्न–भिन्न मान के लिय 'A' का परिसर होगा :

(A) $\left(\frac{5}{2}, \frac{7}{2}\right)$ (B) (4, 5) (C) $\left(4, \frac{9}{2}\right)$ (D) (3, 4)

Ans. D

23. The length of portion of straight line inside the triangle may lie in the range :
इस रेखा का वह हिस्सा जो कि त्रिभुज के भीतर है, उसकी लम्बाई का परिसर होगा :

(A) (2, 4) (B) $\left(\frac{3}{2}, \sqrt{3}\right)$ (C) $(\sqrt{2}, 2)$ (D) $(\sqrt{2}, \sqrt{3})$

Ans. C

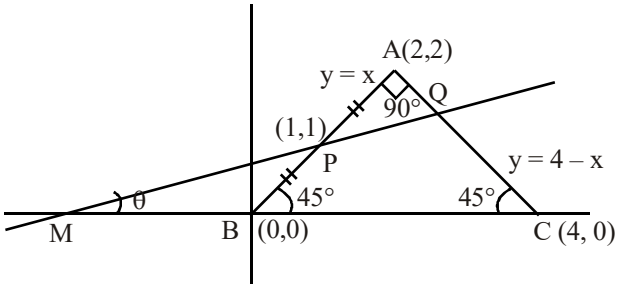
Sol. Equation of line PM :

$y - 1 = \tan\theta\,(x - 1)$

Intersection point 'Q' of AC and MP

$4 - x - 1 = \tan\theta\,(x - 1)$

$Q \equiv \left(\frac{3+\tan\theta}{1+\tan\theta}, \frac{1+3\tan\theta}{1+\tan\theta}\right)$



$$\text{Area of } \Delta APQ = \text{modulus of } \frac{1}{2}\begin{vmatrix} 1 & 1 & 1 \\ 2 & 2 & 1 \\ \frac{3+\tan\theta}{1+\tan\theta} & \frac{1+3\tan\theta}{1+\tan\theta} & 1 \end{vmatrix} = \frac{1-\tan\theta}{1+\tan\theta}$$

MATCH THE COLUMN TYPE QUESTION(S)

24. Match the following column :

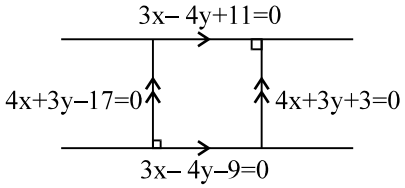
Column–I	**Column–II**
(A) The four lines $3x - 4y + 11 = 0$; $3x - 4y - 9 = 0$; $4x + 3y + 3 = 0$ and $4x + 3y - 17 = 0$ enclose a figure which is	(P) A quadrilateral which is neither a parallelogram nor a trapezium nor a kite.
(B) The lines $2x + y = 1$, $x + 2y = 1$, $2x + y = 3$ and $x + 2y = 3$ from a figure which is	(Q) A parallelogram which is neither a rectangle nor a rhombus
(C) If 'O' is the origin, P is the intersection of the lines $2x^2 - 7xy + 3y^2 + 5x + 10y - 25 = 0$, A and B are the points in which these lines are cut by the line $x + 2y - 5 = 0$, then the points O, A, P, B (in some order) are the vertices of	(R) A rhombus which is not a square
	(S) A square

निम्न कॉलम को सुमेलित कीजिए :

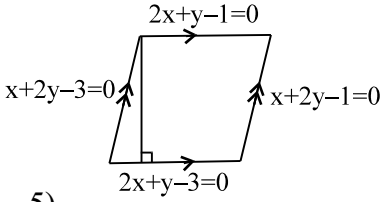
Column–I	**Column–II**
(A) चार भुजाऐं $3x - 4y + 11 = 0$; $3x - 4y - 9 = 0$; $4x + 3y + 3 = 0$ एवं $4x + 3y - 17 = 0$ किस आकृति को बनाएगी।	(P) एक चतुर्भुज जो न तो समान्तर चतुर्भुज एवं नही काइट है
(B) रेखाऐं $2x + y = 1$, $x + 2y = 1$, $2x + y = 3$ तथा $x + 2y = 3$ किस आकृति को बनाएगी	(Q) एक समान्तर चतुर्भुज जो न तो आयत एवं न ही समचतुर्भुज है
(C) यदि रेखा $2x^2 - 7xy + 3y^2 + 5x + 10y - 25 = 0$ का प्रतिच्छेद बिन्दु P हो तथा इस रेखा युग्म को दूसरी रेखा $x + 2y - 5 = 0$ यदि A एवं B पर काटती है तथा O मूलबिन्दु है। तो O, A, P, B किसके शीर्ष होगे	(R) एक समचतुर्भुज जो वर्ग नहीं है
	(S) वर्ग

Ans. (A) S; (B) R; (C) Q

Sol. (A) $\left.\begin{array}{l} d_1 = \dfrac{20}{5} = 4 \\ d_2 = \dfrac{20}{5} = 4 \end{array}\right] \Rightarrow$ square

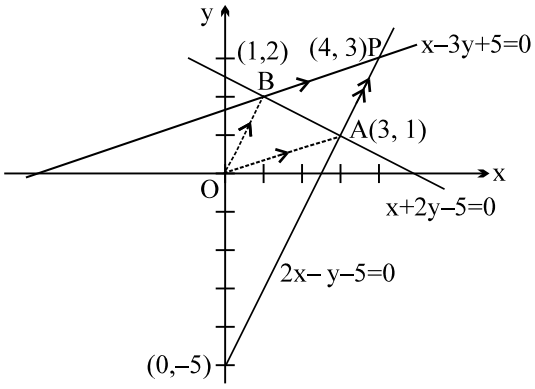


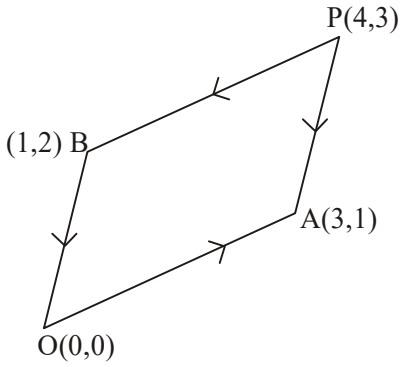
(B) $\left.\begin{array}{l} d_1 = \dfrac{2}{\sqrt{5}} \\ d_2 = \dfrac{2}{\sqrt{5}} \end{array}\right] \Rightarrow$ interior not 90° $\Rightarrow$ rhombus



(C) $2x^2 - 7xy + 3y^2 + 5x + 10y - 25 = 0 \equiv (x - 3y + 5)(2x - y - 5)$

the point of intersection is (4, 3)





$m_{OA} = \dfrac{1}{3}$ $\qquad$ $m_{BP} = \dfrac{1}{3}$

$m_{AP} = \dfrac{2}{1}$ $\qquad$ $m_{OB} = \dfrac{2}{1}$

$OA \parallel AP$ & $AP \parallel OB$

$OA \not\perp AP$

Hence, parallelogram.

25. Consider the 3 linear equations

$ax + by + c = 0$

$bx + cy + a = 0$

$cx + ay + b = 0$

where $a, b, c \in R$.

Column–I	**Column–II**
(A) If $a + b + c = 0$ and $a^2 + b^2 + c^2 \neq ab + bc + ca$ then	(P) Entire xy plane
(B) If $a + b + c = 0$ and $a^2 + b^2 + c^2 = ab + bc + ca$ then	(Q) The lines are concurrent
(C) If $a + b + c \neq 0$ and $a^2 + b^2 + c^2 \neq ab + bc + ca$ then	(R) Line are concident
(D) If $a + b + c \neq 0$ and $a^2 + b^2 + c^2 = ab + bc + ca$ then	(S) Lines are neither coincident nor concurrent

तीन रेखाओं का समीकरण दिया है

$ax + by + c = 0$

$bx + cy + a = 0$

$cx + ay + b = 0$

जहाँ $a, b, c \in R$.

Match the following column :

Column – I	**Column – II**
(A) यदि $a + b + c = 0$ एवं $a^2 + b^2 + c^2 \neq ab + bc + ca$ तो	(P) xy तल होगा
(B) यदि $a + b + c = 0$ एवं $a^2 + b^2 + c^2 = ab + bc + ca$ तो	(Q) रेखाऐं संगामी होगी
(C) यदि $a + b + c \neq 0$ एवं $a^2 + b^2 + c^2 \neq ab + bc + ca$ तो	(R) रेखाऐं सम्पाती होगी
(D) यदि $a + b + c \neq 0$ एवं $a^2 + b^2 + c^2 = ab + bc + ca$ तो	(S) न तो रेखाऐं सम्पाती होगी एवं न ही संगामी होगी

Ans. (A) Q; (B) P; (C) S; (D) R

Sol. $D = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ and $D = 0$ is condition of concurrency

$D = -(a^3 + b^3 + c^3 - 3abc) = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$

(A) if $a + b + c = 0$ but $\sum a^2 \neq \sum ab$ i.e. a, b, c are not all equal, then $D = 0$

hence lines are concurrent $\Rightarrow$ (Q)

(B) if $a + b + c = 0$ and $\sum a^2 = \sum ab$ $\Rightarrow a = b = c$

$\therefore a = 0; \quad b = 0; \quad c = 0$

$\Rightarrow$ lines becomes identical and of the form $0x + 0y + 0 = 0$

any ordered pair (x, y) will satisfy $\Rightarrow$ complete xy plane $\Rightarrow$ (P)

(C) if $a + b + c \neq 0$ and $\sum a^2 \neq \sum ab$ $\Rightarrow$ a, b, c are not all equal $\Rightarrow D \neq 0$

In this case equations represents set of lines which are neither concident nor concurrent $\Rightarrow$ (S)

(D) if $a + b + c \neq 0$ and $\sum a^2 = \sum ab$ $\Rightarrow a = b = c$

hence lines becomes identical or concident $\Rightarrow$ (R)

26. Match the following column :

Column–I	**Column–II**
(A) The lines $y = 0$; $y = 1$; $x - 6y + 4 = 0$ and $x + 6y - 9 = 0$ constitute a figure which is	(P) a cyclic quadraliteral
(B) The points A(a, 0), B(0, b), C(c, 0) and D(0, d) are such that $ac = bd$ and a, b, c, d are all non-zero. The points A, B, C and D always constitute	(Q) a rhombus
(C) The figure formed by the four lines $ax \pm by \pm c = 0$ $(a \neq b)$, is	(R) a square
(D) The line pairs $x^2 - 8x + 12 = 0$ and $y^2 - 14y + 45 = 0$ constitute a figure which is	(S) a trapezium

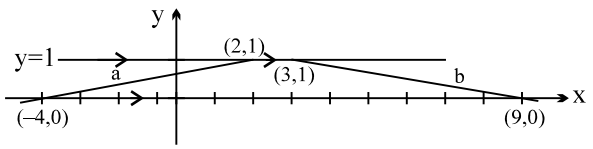
निम्न कॉलम को सुमेलित कीजिए :

Column–I	**Column–II**
(A) रेखा $y = 0$; $y = 1$; $x - 6y + 4 = 0$ एवं $x + 6y - 9 = 0$ से बनने वाली आकृति होगी	(P) चक्रीय चतुर्भुज
(B) यदि $a, b, c, d \neq 0$ तथा A(a, 0), B(0, b), C(c, 0) एवं D(0, d) इस प्रकार से है कि $ac = bd$ है तो आकृति A B C D इनमें से क्या होगा	(Q) समचतुर्भुज
(C) चार रेखाओं $ax \pm by \pm c = 0$ $(a \neq b)$ से बनने वाली आकृति होगी	(R) वर्ग
(D) रेखा युग्म $x^2 - 8x + 12 = 0$ एवं $y^2 - 14y + 45 = 0$ कौनसी आकृति बनाएगें।	(S) समलम्ब चतुर्भुज

Ans. (A) P, S; (B) P; (C) Q; (D) P, Q, R

Sol. (A) Obviously trapezium

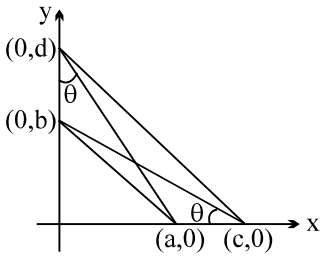
$$\left.\begin{matrix} a = \sqrt{37} \\ b = \sqrt{37} \end{matrix}\right] \quad \Rightarrow \quad a = b$$



hence isosceles trapezium

$\Rightarrow$ a cyclic quadrilateral also $\Rightarrow$ P, S

(B) $ac = bd$ $\Rightarrow$ $\dfrac{b}{c} = \dfrac{a}{d}$



$$\left.\begin{matrix} \tan\theta = \dfrac{b}{c} \\ \tan\phi = \dfrac{a}{d} \end{matrix}\right] \quad \Rightarrow \quad \theta = \phi$$

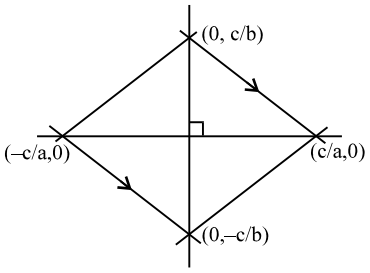
hence cyclic quadrilateral $\Rightarrow$ P

(C) $ax \pm by \pm c = 0$

if $y = 0$, $x = \pm \frac{c}{a}$

if $x = 0$, $y = \pm \frac{c}{b}$

$\Rightarrow$ rhombus $\Rightarrow$ Q



(D) $(x - 6)(x - 2) = 0$

$x = 6$ and $x = 2$

$y^2 - 14y + 45 = 0$

$(y - 9)(y - 5) = 0$

$\Rightarrow$ a square $\Rightarrow$ P, Q, R

